

Assessing Drug Effectiveness And Analyzing Sentiment Using A Drug Review Dataset

**BHUMIKA SAXENA
DEPARTMENT OF HEALTH INFORMATION PROCESSING - BINF 5312
RUTGER'S UNIVERSITY
Dr. LORENA CARLO
SPRING - 2025**

ABSTRACTS

This study explores the application of natural language processing in analyzing patient drug reviews to predict drug effectiveness and classify user sentiments.

Using the druglib.com dataset, various techniques including VADER, BERT, Scispacy, BioBERT were employed for sentiment analysis, Name Entity Recognition, and effectiveness prediction.

Results show a strong correlation between sentiment scores and drug ratings, highlights the role of patient feedback in the real world drug performance assessment.

1.INTRODUCTION

The intersection of healthcare and technology presents significant opportunities to enhance patient care through the analysis of publicly available data. In this project, we will investigate the potential for accurately predicting drug effectiveness and user sentiment by utilizing the Drug Review Dataset from DrugLib.com. User-generated reviews provide critical insights into drug performance, user experiences, and sentiment—assets that are invaluable for pharmaceutical companies, healthcare professionals, and patients. By employing advanced natural language processing (NLP) techniques, we will uncover trends and patterns in user reviews, predict drug ratings, and analyze sentiment. This research question is essential as it underscores the vital role of user feedback in driving improvements in healthcare outcomes, and it clearly demonstrates the promising applications of NLP in healthcare analytics.(Hirschberg & Manning, 2015).

GOALS AND OBJECTIVES

Goals

This study aims to :

- 1. See how well drugs work using patient reviews.**
- 2. Look at feelings (sentiments) people have about different types of drugs.**
- 3. Find out what worries people about side effects and how happy they are.**
- 4. Compare different natural language processing (NLP) methods to find the best way to analyze patient feedback.**

RESEARCH QUESTIONS

- **Can we use sentiment analysis to accurately identify patient experiences as positive, negative, or neutral.**
- **How do patient sentiments scores relate to how effective the drugs are rated.**
- **What common topic appear in patient reviews about drug side effects and benefits.**
- **What NLP technique offers the best accuracy in predicting drug effectiveness?**

3. Dataset Information

The dataset used for this project is the Drug Review Dataset (DrugLib.com), obtained from the UCI Machine Learning Repository ([source](#)).

- Source: [UCI Machine Learning Repository – Drug Review Dataset \(DrugLib.com\)](#)

Key Information About the Dataset

- Where did you find the dataset?
 - It is publicly available on the UCI Machine Learning Repository.
- Why was the data created, and for what purpose?
 - The dataset was compiled to analyze real-world patient experiences with different drugs, focusing on effectiveness, side effects, and satisfaction.
- Who collected the data?
 - The dataset was sourced from DrugLib.com, an online drug review platform where users provide feedback on medications.
- How large is the dataset?
 - The dataset consists of 4,150 patient reviews, containing information such as:
 - Drug Name (Categorical)
 - Condition (Categorical)
 - Review Text (Unstructured)
 - Satisfaction Rating (Numerical, 1-10)
 - Side Effects (Categorical)
- Important publications associated with the dataset
- Prior studies have used the dataset for sentiment analysis, drug effectiveness prediction, and topic modeling in healthcare research.

METHODOLOGY

To effectively achieve goals, proposing and implementing the following Natural Language Processing (NLP) methods:

1. Sentiment Analysis

Purpose:

The primary aim is to classify patient reviews as positive, negative, or neutral. This classification will help us gauge overall patient satisfaction and identify areas for improvement in healthcare services.

Method:

We used VADER (Valence Aware Dictionary and sentiment Reasoner) for initial sentiment scoring.(Zolnoori et al., 2020).

VADER is particularly useful for social media and short text, making it a great choice for patient reviews. - In addition, we implemented BERT-based (Bidirectional Encoder Representations from Transformers) sentiment classification.(Devlin et al., 2019).

This advanced model uses deep learning techniques to better understand context and enhance accuracy in identifying sentiment.

Variables Used:

The analysis will rely on the Review Text, which contains the patients' comments, benefits and side effects which gives an overall score of their experience.

2.Named Entity Recognition (NER)

Purpose:

This method aims to identify and extract relevant medical terms, drug names, specific symptoms, and conditions mentioned in the reviews. Understanding these entities will provide insights into the primary concerns and needs of patients.

Method:

We will utilize ScispaCy, which is a specialized NLP model designed for processing biomedical text.(Alsentzer et al., 2019).

It can accurately extract medical entities from complex sentences. - To further improve accuracy, we will apply transformer-based BioNER (Biomedical Named Entity Recognition) models. These models leverage deep learning to better recognize and classify medical terminology in the reviews.

Variables Used:

We will focus on the Review Text to extract relevant entities and the Condition to analyze specific patient health issues.

3. Text Classification for Drug Effectiveness Prediction

Purpose:

The goal here is to predict drug effectiveness ratings based on the insights gathered from patient reviews. This prediction will help healthcare providers understand how well certain drugs are performing in real-world scenarios.

Method:

First, we will apply TF-IDF (Term Frequency-Inverse Document Frequency) with Logistic Regression as a baseline approach for our analysis. TF-IDF helps in identifying the importance of words in documents, which can be useful for classification. - After establishing a baseline, we will enhance our predictive capabilities by using advanced transformer-based models like BERT and RoBERTa. (Gurulingappa et al., 2012).

These models are known for their superior performance in text classification tasks due to their ability to capture intricate language patterns.

Variables Used:

The analysis will rely on the Review Text to extract sentiments and themes related to drug effectiveness and the Effectiveness Rating to quantify how well these drugs are perceived by patients.

• Steps to perform sentiment Analysis

Import Libraries

```
%%capture
import nltk
import pandas as pd
from sklearn.metrics import classification_report
from nltk.classify import NaiveBayesClassifier
import spacy
import sys
!pip install scispacy
```

Import Dataset

```
Drug_lib_df = pd.read_csv("drugLibTest_NLP Final Project Dataset.csv")
Drug_lib_df.head()
```

Unnamed: 0	urlDrugName	rating	effectiveness	sideEffects	condition	benefitsReview	sideEffectsReview	commentsReview	
0	1366	biaxin	9	Considerably Effective	Mild Side Effects	sinus infection	The antibiotic may have destroyed bacteria cau...	Some back pain, some nauseau.	Took the antibiotics for 14 days. Sinus infect...
1	3724	lamictal	9	Highly Effective	Mild Side Effects	bipolar disorder	Lamictal stabilized my serious mood swings. On...	Drowsiness, a bit of mental numbness. If you t...	Severe mood swings between hypomania and depre...
2	3824	depakene	4	Moderately Effective	Severe Side Effects	bipolar disorder	Initial benefits were comparable to the brand ...	Depakene has a very thin coating, which caused...	Depakote was prescribed to me by a Kaiser psyc...
3	969	sarafem	10	Highly Effective	No Side Effects	bi-polar / anxiety	It controls my mood swings. It helps me think...	I didnt really notice any side effects.	This drug may not be for everyone but its wond...
4	696	accutane	10	Highly Effective	Mild Side Effects	nodular acne	Within one week of treatment superficial acne ...	Side effects included moderate to severe dry s...	Drug was taken in gelatin tablet at 0.5 mg per...

```
print("number of drug_lib transcripts: ", len(Drug_lib_df))
```

number of drug_lib transcripts: 1036

```
%%capture
import sys
!{sys.executable} -m pip install spacy
!{sys.executable} -m pip install scispacy
```

```
from nltk.corpus import stopwords # Stopword removal
from nltk.tokenize import word_tokenize # Tokenization
from nltk.stem import WordNetLemmatizer # Lemmatization
```

Rows With Null Values

```
#Check rows with null values
Drug_lib_df.isnull().sum().sort_values(ascending = False)
```

sideEffectsReview 23
benefitsReview 5
commentsReview 1
Unnamed: 0 0
urlDrugName 0
rating 0
effectiveness 0
sideEffects 0
condition 0
dtype: int64

Removing Row With Null Values

```
#Remove rows with nulls
Drug_lib_df = Drug_lib_df[Drug_lib_df['sideEffectsReview'].notna()]
Drug_lib_df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 1013 entries, 0 to 1035
Data columns (total 9 columns):
#   Column              Non-Null Count  Dtype
---  -
0   Unnamed: 0          1013 non-null  int64
1   urlDrugName         1013 non-null  object
2   rating              1013 non-null  int64
3   effectiveness       1013 non-null  object
4   sideEffects         1013 non-null  object
5   condition           1013 non-null  object
6   benefitsReview      1008 non-null  object
7   sideEffectsReview   1013 non-null  object
8   commentsReview      1012 non-null  object
dtypes: int64(2), object(7)
memory usage: 79.1+ KB
```

```
#Remove rows with nulls
Drug_lib_df = Drug_lib_df[Drug_lib_df['benefitsReview'].notna()]
Drug_lib_df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 1008 entries, 0 to 1035
Data columns (total 9 columns):
#   Column              Non-Null Count  Dtype
---  -
0   Unnamed: 0          1008 non-null  int64
1   urlDrugName         1008 non-null  object
2   rating              1008 non-null  int64
3   effectiveness       1008 non-null  object
```

```
#Remove rows with nulls
Drug_lib_df = Drug_lib_df[Drug_lib_df['commentsReview'].notna()]
Drug_lib_df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 1007 entries, 0 to 1035
Data columns (total 9 columns):
#   Column              Non-Null Count  Dtype
---  -
0   Unnamed: 0          1007 non-null  int64
1   urlDrugName         1007 non-null  object
2   rating              1007 non-null  int64
3   effectiveness       1007 non-null  object
4   sideEffects         1007 non-null  object
5   condition           1007 non-null  object
6   benefitsReview      1007 non-null  object
7   sideEffectsReview   1007 non-null  object
8   commentsReview      1007 non-null  object
dtypes: int64(2), object(7)
memory usage: 78.7+ KB
```

Converting Text into Lowercase

```
#Convert sideEffectsReview into lowercase
Drug_lib_df['sideEffectsReview'] = Drug_lib_df['sideEffectsReview'].str.lower()
Drug_lib_df.head()
```

	Unnamed: 0	urlDrugName	rating	effectiveness	sideEffects	condition	benefitsReview	sideEffectsReview	commentsReview
0	1366	biacin	9	Considerably Effective	Mild Side Effects	sinus infection	The antibiotic may have destroyed bacteria cau...	some back pain, some nauseau.	Took the antibiotics for 14 days. Sinus infect...
1	3724	lamictal	9	Highly Effective	Mild Side Effects	bipolar disorder	Lamictal stabilized my serious mood swings. On	drowsiness, a bit of mental numbness if you t	Severe mood swings between hypomania and depressive

Download Vader Lexicon

```
nltk.download('vader_lexicon')
```

True

Applying VADER Sentiment Analysis on Comments, Benefits, and Side Effects Reviews"

```
from nltk.sentiment.vader import SentimentIntensityAnalyzer
import nltk
nlk.download('vader_lexicon')
```

```
# Initialize VADER
sid = SentimentIntensityAnalyzer()
```

```
def label_sentiment(score):
    if score >= 0.05:
        return 'positive'
    elif score <= -0.05:
        return 'negative'
    else:
        return 'neutral'
```

```
columns_to_check = ['commentsReview', 'benefitsReview', 'sideEffectsReview']
```

```
for col in columns_to_check:
    print(f"\nSentiment analysis for: {col}")
    Drug_lib_df['vader_compound'] = Drug_lib_df[col].astype(str).apply(lambda x: sid.polarity_scores(x)['compound'])
    Drug_lib_df['vader_label'] = Drug_lib_df['vader_compound'].apply(label_sentiment)
    sentiment_counts = Drug_lib_df['vader_label'].value_counts(normalize=True) * 100
    print(sentiment_counts)
```

Sentiment analysis for: commentsReview
vader_label

positive 36.345581

negative 35.451837

neutral 28.202582

Name: proportion, dtype: float64

Sentiment analysis for: benefitsReview
vader_label

positive 46.772592

negative 38.828203

neutral 14.399206

Name: proportion, dtype: float64

Sentiment analysis for: sideEffectsReview
vader_label

negative 54.419067

neutral 25.521351

positive 20.059583

Name: proportion, dtype: float64

Load BERT Tokenizer and Model for Sentiment Analysis

```
] : from transformers import AutoTokenizer, AutoModelForSequenceClassification
    from transformers import TextClassificationPipeline

model_name = "nlptown/bert-base-multilingual-uncased-sentiment"
tokenizer = AutoTokenizer.from_pretrained(model_name)
model = AutoModelForSequenceClassification.from_pretrained(model_name)

from transformers import pipeline

# Load the sentiment analysis pipeline with truncation enabled
sentiment_pipeline = pipeline("sentiment-analysis",
                              model="nlptown/bert-base-multilingual-uncased-sentiment",
                              truncation=True,)

# Create Text Classification Pipeline
health_sentiment = TextClassificationPipeline(model=model, tokenizer=tokenizer, return_all_scores=False)

reviews = Drug_lib_df['benefitsReview'].astype(str).tolist()
sentiment_results = sentiment_pipeline(reviews)

# Extract labels and scores
sentiment_labels = [result['label'] for result in sentiment_results]
sentiment_scores = [result['score'] for result in sentiment_results]
```



```
# Select a subset of benefit reviews
sample_reviews = Drug_lib_df['benefitsReview'].dropna().head(50).tolist()

# Run sentiment prediction
predicted_sentiment = health_sentiment(sample_reviews)

index_values = Drug_lib_df.head(len(predicted_sentiment)).index

# Add star label to DataFrame using index_values
Drug_lib_df.loc[index_values, 'bert_star_label'] = [res['label'] for res in predicted_sentiment]
```

```
def star_to_sentiment(label):
    stars = int(label.split()[0])
    if stars <= 2:
        return 'negative'
    elif stars == 3:
        return 'neutral'
    else:
        return 'positive'

Drug_lib_df.loc[index_values, 'bert_star_label'] = [res['label'] for res in predicted_sentiment]
```

```
Drug_lib_df[['benefitsReview', 'bert_sentiment_label', 'bert_sentiment_score']].head()
```

	benefitsReview	bert_sentiment_label	bert_sentiment_score
0	the antibiotic may have destroyed bacteria cau...	3 stars	0.431390
1	lamictal stabilized my serious mood swings. on...	5 stars	0.633112
2	initial benefits were comparable to the brand ...	4 stars	0.320670
3	it controls my mood swings. it helps me think...	5 stars	0.763892
4	within one week of treatment superficial acne	1 star	0.584040

```
# Function to truncate text and avoid token limit errors
def get_bert_sentiment(text):
    return sentiment_pipeline(text)[0]['label']

# Apply the function to the dataset
Drug_lib_df['benefitsReview_raw'] = Drug_lib_df['benefitsReview'].apply(lambda x: get_bert_sentiment(str(x)))
```

```
# Convert rating (1-5 stars) to positive, neutral, negative
def simplify(label):
    stars = int(label[0])
    if stars >= 4:
        return 'positive'
    elif stars == 3:
        return 'neutral'
    else:
        return 'negative'
```

```
Drug_lib_df['benefitsReview_sentiment'] = Drug_lib_df['benefitsReview_raw'].apply(simplify)
```

```
# Show results
Drug_lib_df[['benefitsReview', 'benefitsReview_raw', 'benefitsReview_sentiment']].head()
```

	benefitsReview	benefitsReview_raw	benefitsReview_sentiment
0	the antibiotic may have destroyed bacteria cau...	3 stars	neutral
1	lamictal stabilized my serious mood swings. on...	5 stars	positive
2	initial benefits were comparable to the brand ...	4 stars	positive
3	it controls my mood swings. it helps me think...	5 stars	positive
4	within one week of treatment superficial acne ...	1 star	negative


```

# Function to truncate text and avoid token limit errors
def get_bert_sentiment(text):
    return sentiment_pipeline(text)[0]['label']

# Apply the function to the dataset
Drug_lib_df['commentsReview_raw'] = Drug_lib_df['commentsReview'].apply(lambda x: get_bert_sentiment(str(x)))

# Convert rating (1-5 stars) to positive, neutral, negative
def simplify(label):
    stars = int(label[0]) # Extract the first digit
    if stars >= 4:
        return 'positive'
    elif stars == 3:
        return 'neutral'
    else:
        return 'negative'

Drug_lib_df['commentsReview_sentiment'] = Drug_lib_df['commentsReview_raw'].apply(simplify)

# Show results
Drug_lib_df[['commentsReview', 'commentsReview_raw', 'commentsReview_sentiment']].head()

```

	commentsReview	commentsReview_raw	commentsReview_sentiment
0	took the antibiotics for 14 days. sinus infect...	1 star	negative
1	severe mood swings between hypomania and depre...	2 stars	negative
2	depakote was prescribed to me by a kaiser psyc...	2 stars	negative
3	this drug may not be for everyone but its wond...	5 stars	positive
4	drug was taken in gelatin tablet at 0.5 mg per...	1 star	negative

Results

- SENTIMENT ANALYSIS RESULT

```
Sentiment analysis for: commentsReview
vader_label
positive    36.345581
negative    35.451837
neutral     28.202582
Name: proportion, dtype: float64
```

```
Sentiment analysis for: benefitsReview
vader_label
positive    46.772592
negative    38.828203
neutral     14.399206
Name: proportion, dtype: float64
```

```
Sentiment analysis for: sideEffectsReview
vader_label
negative    54.419067
neutral     25.521351
positive    20.059583
Name: proportion, dtype: float64
```

	commentsReview	commentsReview_raw	commentsReview_sentiment
0	took the antibiotics for 14 days. sinus infect...	1 star	negative
1	severe mood swings between hypomania and depre...	2 stars	negative
2	depakote was prescribed to me by a kaiser psyc...	2 stars	negative
3	this drug may not be for everyone but its wond...	5 stars	positive
4	drug was taken in gelatin tablet at 0.5 mg per...	1 star	negative

	benefitsReview	bert_sentiment_label	bert_sentiment_score
0	the antibiotic may have destroyed bacteria cau...	3 stars	0.431390
1	lamictal stabilized my serious mood swings. on...	5 stars	0.633112
2	initial benefits were comparable to the brand ...	4 stars	0.320670
3	it controlls my mood swings. it helps me think...	5 stars	0.763892
4	within one week of treatment superficial acne ...	1 star	0.584940

	benefitsReview	benefitsReview_raw	benefitsReview_sentiment
0	the antibiotic may have destroyed bacteria cau...	3 stars	neutral
1	lamictal stabilized my serious mood swings. on...	5 stars	positive
2	initial benefits were comparable to the brand ...	4 stars	positive
3	it controlls my mood swings. it helps me think...	5 stars	positive
4	within one week of treatment superficial acne ...	1 star	negative

To better understand how patients feel about their medications, we applied two sentiment analysis methods—VADER and BERT—across three key review categories: **benefits**, **side effects**, and **general comments**.

VADER Analysis (Rule-Based)

- **Benefits Review:** Nearly half (46.8%) of users described their experience positively, while 38.8% had negative experiences and 14.4% were neutral.
- **Side Effects Review:** Over half (54.4%) of the sentiments were negative, highlighting common dissatisfaction related to side effects.
- **Comments Review:** Sentiment here was more balanced—36.3% positive, 35.5% negative, and 28.2% neutral—showing a wide range of patient experiences.

BERT Analysis (Transformer-Based Model)

We also used a more advanced BERT model to analyze the **benefitsReview** column. BERT generates a star rating (1 to 5) along with a confidence score. These were then grouped into positive, neutral, and negative sentiments for easier interpretation.

Examples:

- A 5-star review like “Lamictal stabilized my serious mood swings...” was labeled as **positive**.
- A 3-star comment (e.g., “The antibiotic may have destroyed bacteria...”) was marked as **neutral**.
- A 1-star review describing poor results was classified as **negative**.

DISCUSSION

Strengths:

- **Comprehensive Sentiment Coverage:**

By combining both VADER and BERT, we captured a broad spectrum of patient opinions—ranging from quick sentiment snapshots to deeply contextual insights. This dual approach helped balance speed with depth.

- **Ease of Interpretation:**

VADER's rule-based system quickly flagged sentiment trends, making it useful for high-level summaries and dashboards. BERT, while more complex, allowed for precise interpretation of longer, nuanced reviews.

- **Real-World Insight:**

The models revealed valuable patterns—patients generally appreciate medication benefits, but side effects remain a significant source of concern. This insight can inform healthcare providers, researchers, and developers aiming to improve patient care and drug design.

Challenges:

- **VADER's Limitations:**

VADER struggles with sarcasm, mixed emotions, and longer text. It treats each sentence individually and lacks the ability to interpret subtle context, which can lead to misclassifications in emotionally rich reviews.

- **BERT's Resource Intensity:**

While BERT excels at context-aware sentiment detection, it requires more processing power and time. It's also more complex to implement and fine-tune, especially when dealing with large datasets.

- **Confidence Score Variability:**

BERT's confidence scores varied widely (0.32 to 0.76), meaning some predictions were less certain. This underscores the importance of cross-checking results or using ensemble methods for higher reliability.

References:

1. Dataset

Dua, D., & Graff, C. (2017). *Drug Review Dataset (DrugLib.com)*. UCI Machine Learning Repository. <https://archive.ics.uci.edu/dataset/461/drug+review+dataset+druglib+com>

2. Liu, B. (2020) – Sentiment Analysis : Mining opinions, sentiments, and emotions (2nd ed.). Cambridge University Press.

3. Sarker, A., Gonzalez, G., & Nikfarjam, A. (2019).– Pharmacovigilance and social media :Utilizing social media data for pharmacovigilance: A review. *Journal of Biomedical Informatics*, 54, 202–212. <https://doi.org/10.1016/j.jbi.2015.12.004>

4. Devlin, J., Chang, M.-W., Lee, K., & Toutanova, K. (2019). BERT: Pre-training of deep bidirectional transformers for language understanding. *arXiv preprint arXiv:1810.04805*. <https://arxiv.org/abs/1810.04805>

5.Alsentzer, E., Murphy, J., Boag, W., Weng, W.-H., Jin, D., Naumann, T., & McDermott, M. (2019). Publicly available clinical BERT embeddings. *Proceedings of the 2nd Clinical Natural Language Processing Workshop*, 72–78. <https://doi.org/10.18653/v1/W19-1909>

6. Gurulingappa, H., Mateen-Rajput, A., & Toldo, L. (2012). Development of a benchmark corpus to support the automatic extraction of drug-related adverse effects. *Journal of Biomedical Informatics*, 45(5), 885–892. <https://doi.org/10.1016/j.jbi.2012.04.008>

7. Zolnoori, M., Fung, K. W., Fontelo, P., Liu, F., & Wu, H. (2020). Sentiment analysis of health-related user reviews using deep learning. *PLOS ONE*, 15(6), e0233990. <https://doi.org/10.1371/journal.pone.0233990>

8 .Hirschberg, J., & Manning, C. D. (2015). Advances in natural language processing. *Science*, 349(6245), 261–266. <https://doi.org/10.1126/science.aaa8685>

9.Miotto, R., Wang, F., Wang, S., Jiang, X., & Dudley, J. T. (2018). Deep learning for healthcare: Review, opportunities and challenges. *Briefings in Bioinformatics*, 19(6), 1236–1246. <https://doi.org/10.1093/bib/bbx044>

10. Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., ... & Duchesnay, É. (2011). Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12, 2825–2830.

11.Zhang, H. (2004). The optimality of Naive Bayes. *Proceedings of the 20th National Conference on Artificial Intelligence*, 3(1), 562–567.